Software Requirements Specification (SRS)

Project: Student Attendance Management System (SAMS)
Version: 1.2
Authors: Kundhan Vudutha (PES1UG24AM327), Tarun J (PES1UG24AM304),
Anish (PES1UG24AM348)
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Revision history

Version	Date	Author	Change summary	Approval
1.0	05-09-2026	Project team	Initial requirements baseline	Pending
1.2	29-10-2026	Current team	Three-member allocation: Developer, Tester and QA; requirements and traceability retained.	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the Student Attendance Management System (SAMS), a web application that records student attendance for each class session and generates attendance reports without manual compilation. It is written for the project guide, the development team, and the evaluators of this course project.

1.2 Scope

SAMS allows faculty to mark attendance for a scheduled class session, allows students to view their own attendance and raise correction requests, and generates automated reports, including a defaulter list for students below a configured minimum-attendance threshold. The system also covers the course, section and timetable setup needed to drive attendance marking, and basic user and role administration.

Out of scope for this version: fee management, examination or grading modules, biometric or facial-recognition hardware integration, and integration with any external college ERP. These may be considered for a future phase.

1.3 Audience

Course faculty, students, the department administrator, the project guide, and the evaluators reviewing this SRS and the resulting implementation.

1.4 Definitions

Acronyms used in this document: SAMS (Student Attendance Management System), UI (User Interface), API (Application Programming Interface), RBAC (Role-Based Access Control), JWT (JSON Web Token), ORM (Object-Relational Mapping), CSV (Comma-Separated Values), SMTP (Simple Mail Transfer Protocol), WCAG (Web Content Accessibility Guidelines).

2. Overall description

2.1 Product perspective

SAMS is a standalone web application built on a Python/Django backend (Django REST Framework) with a PostgreSQL database, served to a browser-based frontend over HTTPS. It does not replace any existing college system in this phase, but its REST API is structured so that a future version could exchange data with an institutional ERP.

2.2 Major product functions (detailed)

- User authentication and role-based access for Student, Faculty and Admin
- Course, section and weekly timetable setup
- Attendance marking per scheduled session, with Present / Absent / On Duty status
- Attendance correction and leave-adjustment workflow with faculty approval
- Automated reports: per-student, per-course, date-range summary, and a defaulter list
- Email and in-app notifications for low attendance and pending correction requests

- Administration of user accounts, roles and the minimum-attendance threshold
- Audit log of every attendance edit

2.3 User roles and characteristics

Student: views their own attendance and course schedule, and may request a correction. Assumed to have only basic computer literacy.

Faculty: marks attendance for the sessions they teach and reviews or approves correction requests. Needs a fast, low-friction marking screen usable between classes.

Admin: sets up courses, timetables and users, configures the attendance threshold, and reviews system-wide reports and the audit log.

2.4 Operating environment

Backend: Python 3.x with Django and Django REST Framework, running behind Gunicorn and Nginx on a Linux server or cloud VM. Database: PostgreSQL 14 or later. Frontend: any current browser (Chrome, Firefox, Edge, Safari) on desktop or mobile, with a minimum supported viewport width of 360px.

2.5 Constraints

- The project is delivered within a single academic semester by a small student team, so scope is limited to the functions listed in 2.2.
- The system must run on modest lab hardware or a free/low-cost cloud tier and use only open-source components.
- Student data handled by the system is subject to the institution's data-privacy policy and must not be exposed to unauthenticated users.

3. External interface requirements

3.1 User interfaces

A responsive web UI provides separate dashboards for Student, Faculty and Admin roles. The faculty attendance-marking screen must allow a full class roll call to be recorded in under one minute. Core screens follow WCAG 2.1 AA colour-contrast guidance and support keyboard navigation.

3.2 Hardware interfaces

A standard desktop, laptop or mobile device with a browser and network access is sufficient for version 1. A classroom webcam or QR/barcode scanner for faster check-in is a possible hardware addition in a later phase but is not required for this release.

3.3 Software interfaces

- Django REST Framework API (JSON over HTTPS) between the frontend and backend
- PostgreSQL database accessed through the Django ORM
- SMTP service (institutional mail server or a provider such as SendGrid) for email notifications
- CSV and PDF export for generated reports

3.4 Communications

All client-server traffic uses HTTPS with TLS 1.2 or later. API calls are authenticated with session cookies or JWTs, depending on the client. Nightly report generation and notification dispatch run as scheduled background jobs (Celery or Django management commands).

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow SAMS-F-###.

4.1 Authentication & Access Control

Description: authenticate users and enforce role-based access to system functions.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SAMS-F-001	The system shall authenticate users using a unique institute ID and password before granting access.	Functional	High	Faculty / Admin	AC-001: Valid credentials load the correct role dashboard; invalid credentials show an error and no session is created. Test: TC-Auth-01	Requires hashed password store
SAMS-F-002	The system shall assign each user exactly one role (Student, Faculty or Admin) and restrict screens and actions to that role.	Functional	High	Admin	AC-002: A Student account cannot open faculty attendance-marking or admin screens. Test: TC-Auth-02	RBAC middleware
SAMS-F-003	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts and log the event.	Functional	Medium	Security	AC-003: The 6th failed attempt returns a lockout message; login is rejected until the timeout expires. Test: TC-Auth-03	–
SAMS-F-004	The system shall let a user reset a forgotten password through a verified email link.	Functional	Medium	Student / Faculty	AC-004: The reset link expires after 30 minutes and can be used once. Test: TC-Auth-04	Requires SMTP integration

4.2 Attendance Marking

Description: record attendance for a scheduled class session and let students view or contest it.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SAMS-F-005	The system shall display the roll list for a faculty member's scheduled session, based on the timetable, on the date of the class.	Functional	High	Faculty	AC-005: Only students enrolled in that section appear; the list matches the timetable slot. Test: TC-Att-01	Requires timetable module
SAMS-F-006	The system shall let faculty mark each student Present, Absent or On Duty for a session from one screen, saved in under one minute for a 60-student class.	Functional	High	Faculty	AC-006: All statuses save correctly; the save action completes within 2 seconds server-side. Test: TC-Att-02	–
SAMS-F-007	The system shall prevent attendance from being marked twice for the same session unless an explicit authorised edit is performed.	Functional	High	Faculty	AC-007: Re-opening a submitted session shows the existing marks and requires an explicit edit action, which is logged. Test: TC-Att-03	Audit log
SAMS-F-008	The system shall let a student view their own attendance percentage per course, updated within 5 minutes of faculty submission.	Functional	High	Student	AC-008: The student dashboard reflects the latest submitted session within the target window. Test: TC-Att-04	–
SAMS-F-009	The system shall let a student raise an attendance correction request with a reason, routed to the concerned faculty for approval or rejection.	Functional	Medium	Student	AC-009: Faculty sees pending requests and can approve or reject; an approved request updates the record and is logged. Test: TC-Att-05	Notification module

4.3 Reports & Analytics

Requirements: the system shall generate per-student, per-course and date-range attendance reports, and a defaulter list of students below a configurable minimum-attendance threshold, exportable as CSV or PDF. IDs SAMS-F-010–SAMS-F-013.

4.4 Course & Timetable Management

Requirements: Admin shall be able to create courses and sections, build a weekly timetable, and assign faculty to sections; faculty attendance screens are generated from this timetable. IDs SAMS-F-014–SAMS-F-016.

4.5 Notifications

Requirements: the system shall email or in-app notify a student when attendance falls below the configured threshold, and notify faculty of pending correction requests. IDs SAMS-F-017–SAMS-F-018.

4.6 Administration & Audit

Requirements: Admin shall manage user accounts and roles, configure the minimum-attendance threshold, and view an audit trail of every attendance edit with timestamp and editor identity. IDs SAMS-F-019–SAMS-F-021.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs SAMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
SAMS-NF-001	Dashboard and attendance-marking pages shall render within 3 seconds on a typical college network for 95% of requests under normal load.	Performance	High	Load test shows 95th percentile ≤ 3s. Test: TC-Perf-01
SAMS-NF-002	The system shall be available during college working hours (8am–6pm, Mon–Sat) with at least 99% uptime, excluding announced maintenance.	Reliability	High	Uptime monitoring report over one semester. Test: Ops log review
SAMS-NF-003	Passwords shall be stored using a salted hash and never in plaintext; all traffic shall use HTTPS.	Security / Compliance	High	Code review and TLS configuration check. Test: TC-Sec-01
SAMS-NF-004	Attendance edit history shall be retained for at least the current academic year for audit purposes.	Audit / Data Retention	Medium	Audit log query returns edits from the full academic year. Test: TC-OPS-01
SAMS-NF-005	The web UI shall be usable from 360px mobile width to desktop widths and follow WCAG 2.1 AA colour-contrast guidance on key screens.	Usability / Accessibility	Medium	Responsive layout check and contrast audit. Test: TC-UX-01
SAMS-NF-006	The system shall support at least 2,000 student and 100 faculty accounts without a change in architecture.	Scalability	Medium	Load test with seeded data at target scale. Test: TC-Perf-02

5.1 Security

5.1.1 Security Objectives

- Protect student personal and academic data from unauthorised access or disclosure.
- Ensure that attendance records, once submitted, cannot be silently altered without a traceable audit trail.
- Restrict every action to the minimum access needed for a user's role (least privilege).

5.1.2 Security Requirements

Req ID	Requirement (shall…)	Type	Priority	Acceptance criteria / Test case ref
SAMS-SR-001	TLS 1.2 or later shall be mandatory for all client-server traffic.	Security	High	AC: server rejects plain HTTP; certificate is valid. Test: TC-Sec-02
SAMS-SR-002	Passwords shall be hashed with a salted algorithm (bcrypt or equivalent) and never logged or transmitted in plaintext.	Security	High	AC: password field is absent from logs and API responses. Test: TC-Sec-03
SAMS-SR-003	Session tokens shall expire after a configurable inactivity period (default 30 minutes) and be invalidated on logout.	Security	High	AC: an expired token is rejected by the API; logout invalidates the token. Test: TC-Sec-04
SAMS-SR-004	Role-based access control shall be enforced server-side on every API endpoint, not only in the UI.	Security	High	AC: a direct API call from a Student token to an Admin-only endpoint is rejected with 403. Test: TC-Sec-05
SAMS-SR-005	Every attendance edit shall be logged with editor ID, timestamp and previous value, and the log shall be append-only.	Security	Medium	AC: an edit produces a log entry; no update or delete path for the log is exposed via the API. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

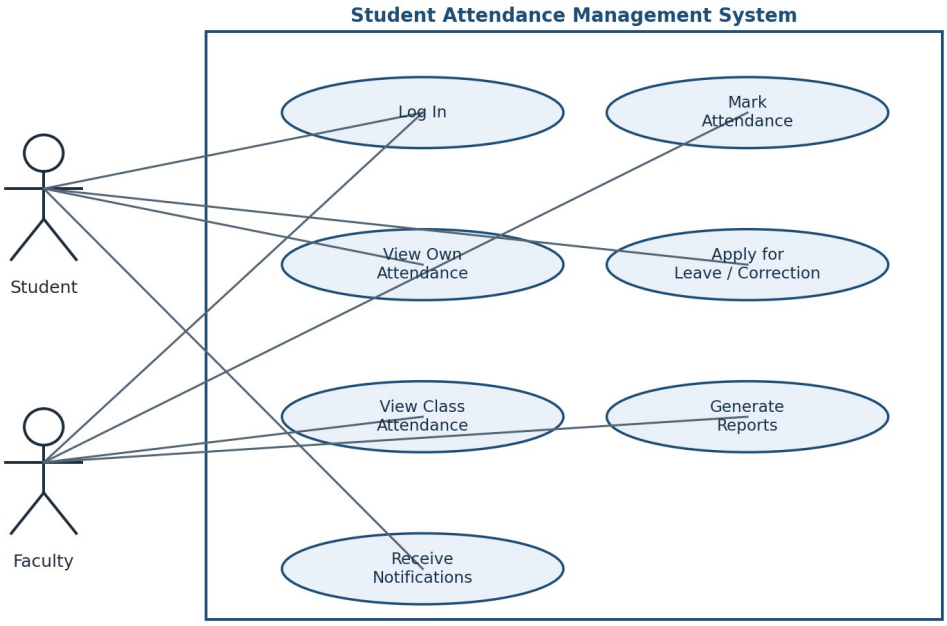
Exit criteria for acceptance: all High-priority functional requirements are implemented and verified, no critical NFR or security-requirement failures remain open, and the RTM shows every listed test case passed or formally waived.

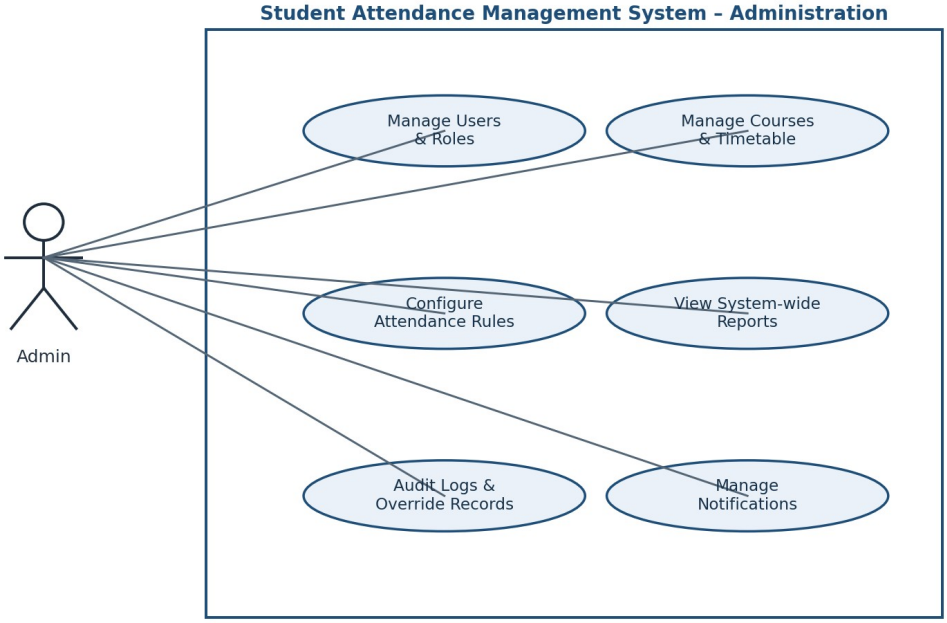
Acceptance test suites: Authentication & Access Control, Attendance Marking, Reports, Notifications, Performance, Security, and Accessibility.

7. System models and diagrams

7.1 UML Use-case diagrams

Figure 1 covers the Student and Faculty interactions with SAMS. Figure 2 covers the Admin interactions.





8. Requirements Traceability Matrix (RTM)

Status key: N = Not started, P = In progress, A = Achieved / passed.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SAMS-F-001	Authenticate user	4.1 / SAD 3.4, 4.3.1, Fig. 7	AuthModule	TC-Auth-01: UT_AUTH_01, IT_AUTH_01, ST_AUTH_01	N	
SAMS-F-003	Lockout after failed attempts	4.1 / SAD 3.4, 3.9, Fig. 7	AuthModule	TC-Auth-03: UT_AUTH_02 to UT_AUTH_04	N	
SAMS-F-006	Mark attendance	4.2 / SAD 4.3.5, Fig. 5	AttendanceModule	TC-Att-02: UT_ATT_03, IT_ATT_01, ST_ATT_01	N	
SAMS-F-007	Prevent duplicate marking	4.2 / SAD 4.3.6, ADR-06	AttendanceModule	TC-Att-03: UT_ATT_04, UT_ATT_07	N	
SAMS-F-009	Attendance correction request	4.2 / SAD 4.3.8, Fig. 6	AttendanceModule / NotifyModule	TC-Att-05: UT_COR_01 to UT_COR_06, ST_COR_01	N	
SAMS-F-013	Defaulter list report	4.3 / SAD 4.3.9	ReportModule	TC-Rep-01: UT_REP_01, UT_REP_02, ST_REP_01	N	
SAMS-F-016	Timetable-driven roll list	4.4 / SAD 4.3.4	CourseModule	TC-Att-01: UT_ATT_01, UT_CRS_02 to UT_CRS_04	N	
SAMS-NF-001	Page load time target	5 / SAD 3.1, 3.7	WebUI / API	TC-Perf-01: ST_PERF_01	N	
SAMS-NF-003	Password hashing & TLS	5 / SAD 3.9	AuthModule / Infra	TC-Sec-01: ST_SEC_01, ST_SEC_02	N	
SAMS-SR-004	Server-side RBAC enforcement	5.1.2 / SAD 3.9, 4.3	API Gateway	TC-Sec-05: IT_AUTH_02, ST_SEC_04	N	